

OSLO GARDERMOEN, Norway

WMO: 013840

Lat: 60.206N

Lon: 11.080E

Elev: 681

StdP: 14.34

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.9	2.8	-5.4	4.2	-1.3	-0.8	5.4	3.6	24.0	36.5	21.0	34.9	2.4	20	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	79.2	61.8	75.8	60.7	72.5	59.1	64.5	73.4	62.8	71.4	61.2	69.3	7.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	82.9	66.3	59.6	78.1	65.3	58.0	73.7	63.8	29.9	73.4	28.6	71.1	27.5	69.6	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	20.5	17.6	15.3	DB	-6.7	82.7	5.5	3.7	-10.7	85.4	-13.9	87.5	-16.9	89.5	-20.9	92.2
(5)				WB	-7.1	66.7	5.4	2.2	-11.0	68.3	-14.2	69.6	-17.2	70.9	-21.1	72.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	23.0	23.9	30.9	40.3	49.9	57.1	62.1	59.7	51.9	41.5	32.6	25.3	
	DBStd	15.60	10.00	9.82	7.93	5.97	6.76	5.03	4.95	4.83	5.35	6.45	7.54	10.02	
	HDD50	4145	836	730	593	296	86	4	0	0	41	269	523	767	
	HDD65	8586	1301	1150	1058	740	471	243	121	178	392	727	973	1232	
	CDD50	1083	0	0	0	5	81	217	374	300	99	7	0	0	
	CDD65	50	0	0	0	0	1	6	30	13	0	0	0	0	
	CDH74	498	0	0	0	0	26	87	283	102	0	0	0	0	
	CDH80	63	0	0	0	0	1	10	46	6	0	0	0	0	
Wind	WSAvg	6.4	5.6	6.0	6.6	6.9	7.4	7.3	6.5	6.0	6.4	6.1	6.1	5.6	
Precipitation	PrecAvg	34.1	2.4	1.9	1.9	1.9	2.6	3.1	3.2	3.8	3.5	3.8	3.5	2.5	
	PrecMax	44.1	6.2	5.7	5.3	4.0	6.1	6.9	8.2	8.7	8.3	8.5	11.8	6.1	
	PrecMin	22.9	0.2	0.2	0.1	0.0	0.0	0.4	0.3	0.2	0.9	0.4	0.6	0.5	
	PrecStd	5.6	1.4	1.2	1.2	1.1	1.4	1.6	1.6	2.0	1.7	2.2	2.1	1.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	44.7	56.4	65.6	77.6	81.0	84.5	80.5	70.6	59.0	51.8	46.4	
		MCWB	39.3	37.8	45.0	50.5	59.2	61.2	63.6	61.8	59.3	52.0	48.0	44.1	
	2%	DB	39.8	40.9	49.5	60.6	71.8	76.5	80.7	77.1	66.6	55.5	48.4	43.2	
		MCWB	37.5	36.4	40.8	48.1	55.5	59.0	62.7	61.8	56.1	51.1	46.2	41.4	
	5%	DB	37.6	38.8	44.7	55.7	67.6	71.8	77.1	73.4	63.8	53.5	45.8	40.7	
		MCWB	35.8	35.9	37.7	45.3	53.7	56.7	61.9	60.5	55.2	49.7	43.9	38.9	
	10%	DB	35.6	36.5	41.1	51.8	62.9	68.0	73.4	69.6	60.7	51.5	43.0	37.8	
		MCWB	34.0	34.5	35.8	43.2	51.3	55.2	60.2	58.8	54.3	48.4	41.4	36.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.6	40.0	45.6	53.0	61.2	63.2	67.1	65.7	61.8	54.9	49.5	44.7
			MCDB	41.6	42.2	55.0	62.1	71.9	73.6	76.7	73.3	66.2	57.3	50.5	45.5
2%		WB	38.1	37.8	41.6	49.2	57.6	61.0	65.3	63.9	59.1	52.8	46.5	41.8	
		MCDB	39.1	39.3	47.5	57.7	68.0	71.5	75.4	71.3	63.8	54.6	47.7	42.8	
5%		WB	36.0	36.3	39.0	46.8	55.3	58.9	63.6	62.3	57.2	50.9	44.0	38.9	
		MCDB	37.1	37.9	42.8	53.7	63.3	68.6	73.1	69.6	61.4	52.8	45.2	40.0	
10%		WB	34.0	34.5	36.8	44.3	52.9	57.0	61.8	60.8	55.5	48.9	41.4	36.3	
		MCDB	35.0	36.1	40.1	50.1	61.8	65.6	70.7	67.7	59.4	50.8	42.6	37.5	
Mean Daily Temperature Range	5% DB	MDBR	9.5	11.7	14.5	16.2	17.5	16.5	16.6	15.9	14.4	11.2	8.4	9.4	
		MCDBR	10.0	11.8	18.7	24.2	25.4	23.8	23.5	21.5	18.4	12.8	11.4	10.5	
		MCWBR	9.0	9.2	12.0	13.6	12.3	10.8	9.7	9.5	10.5	9.4	10.4	9.9	
	5% WB	MCDBR	9.1	10.1	16.6	20.7	21.3	19.4	19.7	17.8	14.7	10.6	10.5	9.8	
		MCWBR	8.8	8.6	11.3	12.4	11.1	10.1	9.3	9.3	9.6	9.2	10.1	9.8	
Clear-Sky Solar Irradiance	taub	0.252	0.295	0.322	0.365	0.363	0.353	0.373	0.365	0.333	0.319	0.272	0.224		
	taud	2.195	2.274	2.318	2.315	2.403	2.452	2.413	2.430	2.511	2.457	2.277	2.102		
	Ebn at Noon	167	224	256	264	274	279	269	262	253	216	159	127		
	Edh at Noon	13	21	28	33	33	32	33	30	23	18	12	9		
All-Sky Solar Radiation	RadAvg	88	285	751	1192	1532	1752	1568	1232	814	360	118	57		
	RadStd	16	52	80	158	158	137	206	104	95	70	33	5		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	+3.65	+3.47	N/A	N/A	N/A	-319	N/A	N/A	N/A
(47)	Regional (56 neighbors)	N/A	N/A	+2.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page